

Login Page (UI) Specification

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Overview

This document specifies the design and functional requirements for the Login Page UI component. The login page is the entry point for users to authenticate to the system using email/username and password credentials.

Page Components

1. Header Section

- Logo (left-aligned)
- Page title: "Sign In"
- Optional tagline: "Welcome back"

2. Login Form

- **Email/Username field**
 - Label: "Email or Username"
 - Type: Text input
 - Placeholder: "Enter your email or username"
 - Required: Yes
 - Validation: Must be non-empty
- **Password field**
 - Label: "Password"
 - Type: Password input (masked characters)
 - Placeholder: "Enter your password"
 - Required: Yes
 - Validation: Must be non-empty
- **"Remember Me" Checkbox**
 - Label: "Remember me on this device"
 - Default: Unchecked
 - Optional: Yes
- **"Forgot Password?" Link**
 - Text: "Forgot password?"
 - Color: Blue (#0066CC)
 - Action: Redirect to password recovery page

3. Submit Button

- Label: "Sign In"
- Type: Primary button (blue, full width)
- State: Enabled when both email and password are filled
- On click: Submit form to `/api/v1/auth/login`
- Loading state: Show spinner and disable button during request

4. Error Message Display

- Location: Above the email field
- Styling: Red background (#FF0000), white text
- Messages to display:
 - "Invalid email or password" (for 401)

- "All fields are required" (for 400)
- "Network error, please try again" (for network failures)
- "Account locked due to too many login attempts" (for rate limiting)

5. Success Redirect

- On successful 200 response: Store JWT token in localStorage/sessionStorage
- Redirect to dashboard or home page
- Clear form fields

6. Footer (Optional)

- Link: "Don't have an account? Sign up"
- Text: "Don't have an account? [Sign up]"
- Color: Gray text with blue link

Functional Requirements

1. **Form Validation**

- Email field: Validate email format (RFC 5322)
- Password field: Min 6 characters
- Show validation errors in real-time (on blur)

2. **API Integration**

- Endpoint: `POST /api/v1/auth/login`
- Request body: `{ "username": "...", "password": "..." }`
- Response: `{ "token": "jwt\_string" }` (200 OK)
- Error responses: 400, 401, 403, 500

3. **Security**

- Never log credentials to console or network tab
- Use HTTPS only
- Implement CSRF protection
- Password field must be masked
- Clear sensitive data on logout

4. **Accessibility**

- All form fields must have `` elements
- Color contrast ratio $\geq 4.5:1$ (WCAG AA)
- Tab order: Email -> Password -> Remember Me -> Submit -> Forgot Password -> Sign Up
- ARIA labels for screen readers

5. **Responsive Design**

- Mobile (320px): Single column, full-width form
- Tablet (768px): Centered form, max-width 400px
- Desktop (1024px): Centered form, max-width 450px

Visual Design

Property	Value
Background Color	#F5F5F5 (light gray)
Form Background	FFFFFF (white)
Primary Button Color	#0066CC (blue)
Error Text Color	CC0000 (red)
Border Radius	4px
Font Family	Arial, sans-serif
Font Size (body)	14px
Font Size (heading)	24px

Test Cases

TC-001: Successful Login

- ****Preconditions****: User has valid credentials
- ****Steps****: Enter email, enter password, click "Sign In"
- ****Expected****: Page redirects to dashboard, JWT stored

TC-002: Invalid Credentials

- ****Preconditions****: User enters wrong password
- ****Steps****: Enter email, enter wrong password, click "Sign In"
- ****Expected****: Error message "Invalid email or password" displayed

TC-003: Missing Email

- ****Preconditions****: Password is filled
- ****Steps****: Leave email empty, click "Sign In"
- ****Expected****: "All fields are required" error shown

TC-004: Missing Password

- ****Preconditions****: Email is filled
- ****Steps****: Leave password empty, click "Sign In"
- ****Expected****: "All fields are required" error shown

TC-005: Remember Me Functionality

- ****Preconditions****: User enters credentials and checks "Remember Me"
- ****Steps****: Log in successfully, close browser, return to site
- ****Expected****: User is still logged in (token in storage)

TC-006: Forgot Password Link

- ****Preconditions****: User on login page
- ****Steps****: Click "Forgot password?" link
- ****Expected****: Redirect to password recovery page

TC-007: Network Timeout

- ****Preconditions****: Server is slow/unavailable
- ****Steps****: Enter credentials, click "Sign In", wait
- ****Expected****: After 5s timeout, show "Network error" message

TC-008: Account Locked

- ****Preconditions****: User made 5+ failed login attempts
- ****Steps****: Try to login again
- ****Expected****: 403 response, show "Account locked" message

Acceptance Criteria

- ☐ Form displays correctly on mobile, tablet, desktop
- ☐ Email/password validation works client-side and server-side
- ☐ Submit button is disabled until both fields are filled
- ☐ API errors are displayed clearly to user
- ☐ JWT token is stored securely after login
- ☐ Forgot password link redirects correctly

- ☐ Sign up link redirects to registration page
- ☐ WCAG AA accessibility compliance
- ☐ All test cases (TC-001 to TC-008) pass